Evaluating the Robustness of Transformer-Based Time-Series Models for Short-Term Financial Volatility Prediction Across Market Regimes

Ziad Tabbouche

Student

***Abstract*— Financial volatility forecasting plays a central role in portfolio optimization, derivative pricing, risk management, and algorithmic trading. Traditional econometric approaches such as GARCH models have long been used for volatility estimation; however, the nonlinear and non-stationary nature of modern financial markets has encouraged the adoption of deep learning methods. Among these, Transformer-based architectures have emerged as promising alternatives for time-series forecasting due to their ability to model long-range temporal dependencies through self-attention mechanisms. Despite their growing popularity, concerns remain regarding their robustness across varying market regimes, particularly during periods of financial instability and structural market shifts. This paper evaluates the robustness of Transformer-based time-series models for short-term financial volatility prediction under distinct market conditions, including bullish, bearish, high-volatility, and crisis-driven regimes. The study compares several Transformer architectures, including Informer, Autoformer, PatchTST, and ETSformer, against benchmark deep learning and econometric models. Experimental results indicate that Transformer models outperform conventional recurrent networks and GARCH-family methods in stable market environments but demonstrate varying levels of robustness during extreme volatility events. PatchTST and ETSformer exhibit superior adaptability across regime transitions due to improved temporal decomposition and long-range dependency modeling. However, model sensitivity to non-stationarity and regime shifts remains a significant limitation. The findings suggest that future volatility forecasting systems should integrate regime-awareness and adaptive attention mechanisms to improve reliability in rapidly evolving financial markets.**

***Index Terms*— Financial Volatility Forecasting, Transformer Models, Time-Series Analysis, Market Regimes, Deep Learning, Financial Risk Prediction, PatchTST, Informer, Autoformer, ETSformer, Quantitative Finance, Artificial Intelligence in Finance**

1. Introduction

Forecasting financial market volatility has remained one of the most important challenges in quantitative finance. Accurate short-term volatility prediction is essential for derivative pricing, portfolio allocation, high-frequency trading, risk management, and market surveillance. Financial institutions rely heavily on volatility forecasts to estimate future risk exposure and optimize investment decisions under uncertainty.

Classical econometric approaches such as Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized ARCH (GARCH) models have traditionally dominated volatility forecasting research. Although these models provide statistical interpretability and strong theoretical foundations, they often struggle to capture nonlinear interactions, structural breaks, and rapidly changing market conditions. Modern financial markets are increasingly characterized by non-stationarity, sudden regime shifts, and high-frequency information flows, limiting the effectiveness of fixed parametric approaches.

Recent advances in deep learning have introduced new opportunities for financial time-series modeling. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) demonstrated improved capacity for modeling nonlinear dependencies in financial data. However, these architectures often encounter difficulties when learning long-range temporal relationships and adapting to dynamic market structures.

Transformer architectures, originally developed for natural language processing, have recently gained significant attention in time-series forecasting tasks. Their self-attention mechanisms enable parallel processing and efficient extraction of long-range dependencies, making them particularly attractive for complex financial sequences. Several specialized Transformer variants, including Informer, Autoformer, PatchTST, and ETSformer, have been proposed to improve forecasting efficiency and scalability in temporal data analysis.

Despite their growing adoption, important questions remain regarding the robustness of Transformer-based volatility forecasting systems under different market regimes. Models trained during stable market conditions may fail during periods of crisis, extreme volatility, or structural transition. Therefore, evaluating model

robustness across varying market environments is critical before deploying these systems in real-world financial applications.

This paper presents a comparative evaluation of Transformer-based time-series models for short-term financial volatility prediction across multiple market regimes. The study investigates model performance, generalization capability, and robustness under changing financial conditions.

2. Literature Review

The application of Transformer architectures to time-series forecasting has expanded rapidly in recent years. Wen et al. provided one of the earliest comprehensive surveys of Transformers in time-series analysis, emphasizing their ability to capture long-range temporal dependencies and complex nonlinear patterns.

Souto and Moradi critically evaluated Transformer-based models for realized volatility forecasting and found that advanced Transformer variants often outperform conventional forecasting approaches under several financial scenarios. Their work highlighted the importance of temporal attention mechanisms in modeling market uncertainty.

Recent research has increasingly focused on regime-sensitive volatility prediction. A study published in the *Journal of Risk and Financial Management* demonstrated that Transformer architectures can detect volatility spikes and market instability more effectively than traditional econometric models, especially during periods of abrupt market transition.

Ramos-Pérez et al. proposed the Multi-Transformer architecture for volatility forecasting and showed that combining multiple attention layers improves predictive accuracy during financial crises such as the COVID-19 pandemic and the 2008 financial collapse.

Another important contribution introduced a multi-transformer neural architecture integrating Transformer layers with GARCH and LSTM models for financial risk assessment. The study demonstrated that hybrid Transformer systems improve forecasting stability across high- and low-volatility periods.

Recent work on volatility-aware temporal Transformers further emphasized the need for regime-sensitive attention mechanisms. Volatility gating modules were proposed to dynamically adapt attention weights according to prevailing market conditions, improving robustness during highly turbulent trading periods.

However, community discussions within machine learning research forums continue to debate whether Transformers consistently outperform simpler models in financial forecasting tasks. Several researchers argue that Transformers may suffer from overfitting, insufficient inductive bias, and poor temporal encoding under non-stationary market conditions.

These concerns highlight the importance of evaluating Transformer robustness across multiple market regimes rather than relying solely on average forecasting accuracy.

3. Methodology

3.1 Dataset Description

The study utilizes historical financial market data collected from major equity indices and financial instruments, including:

- S&P 500
- NASDAQ Composite
- FTSE 100

- EURO/USD Exchange Rate
- Reliance Industries Ltd.
- Samsung Electronics

The dataset includes:

- Open prices
- Close prices
- Trading volume
- Realized volatility
- Intraday return series

The data spans multiple market conditions, including stable growth periods, crisis events, and post-crisis recoveries.

3.2 Market Regime Classification

To evaluate robustness under different financial conditions, the dataset is segmented into distinct market regimes:

- Bull market regime
- Bear market regime
- High-volatility regime
- Low-volatility regime
- Crisis regime
- Recovery regime

Regime classification is performed using volatility thresholds, moving-average indicators, and hidden Markov state identification methods.

3.3 Transformer-Based Models

The following Transformer architectures are evaluated:

- Informer
- Autoformer
- PatchTST
- ETSformer
- Vanilla Transformer Encoder

Benchmark comparison models include:

- GARCH
- EGARCH
- LSTM
- CNN-LSTM
- GRU

The self-attention mechanism central to Transformer architectures is represented by:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

Where:

- **Q** = Query matrix
- **K** = Key matrix
- **V** = Value matrix
- QK^T = Query multiplied by the transpose of Key (measures similarity between queries and keys)
- d_k = dimension of the key vectors (used to scale the values so they don't get too large)
- **softmax(...)** = converts the scaled similarity scores into probabilities (weights that sum to 1)
- The result is multiplied by **V** to produce a weighted sum of the values

3.4 Evaluation Metrics

Model performance is evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- Quasi-Likelihood Loss (QLIKE)
- Directional Accuracy
- Regime Transition Stability

Cross-regime generalization testing is also conducted to assess robustness during unseen market conditions.

4. Experimental Results

4.1 Performance During Stable Market Regimes

Transformer-based architectures achieved strong predictive performance during stable market periods characterized by low volatility and gradual trend evolution. PatchTST and Autoformer demonstrated superior long-range temporal modeling compared with recurrent neural networks.

Patch-based temporal segmentation improved feature extraction efficiency and reduced prediction instability over longer forecasting horizons. These findings are consistent with recent studies showing that Transformers can effectively capture persistent temporal dependencies in financial sequences.

Traditional GARCH models produced competitive short-term forecasts under low-volatility conditions but struggled to capture nonlinear market dynamics compared with Transformer-based systems.

4.2 Performance During High-Volatility Regimes

During periods of elevated market uncertainty, model robustness varied significantly across architectures. Informer and vanilla Transformer models exhibited reduced forecasting stability under abrupt volatility spikes.

ETSformer achieved comparatively stronger performance due to its decomposition-based architecture and frequency-aware attention mechanisms. The integration of exponential smoothing components improved adaptability during rapid market transitions.

Hybrid Transformer-LSTM systems also demonstrated improved resilience during crisis periods by combining sequential memory mechanisms with global attention structures.

4.3 Cross-Regime Generalization

Cross-regime testing revealed that several Transformer architectures experienced performance degradation when evaluated on market conditions not represented during training.

PatchTST demonstrated relatively strong generalization capability due to its efficient temporal representation strategy. However, all evaluated models remained sensitive to severe structural breaks and unexpected financial shocks.

The findings suggest that regime-awareness and adaptive feature selection remain critical for improving forecasting reliability under changing market conditions.

5. Discussion

The experimental results indicate that Transformer-based time-series models provide substantial advantages for short-term financial volatility forecasting, particularly in stable and moderately dynamic environments. Their self-attention mechanisms enable efficient modeling of long-range dependencies and complex temporal interactions that are difficult for recurrent architectures to capture.

Nevertheless, the study also highlights several important limitations.

First, financial markets are inherently non-stationary. Transformer architectures often assume stable temporal relationships during training, reducing robustness when sudden market disruptions occur. Structural breaks, geopolitical events, liquidity crises, and macroeconomic shocks can significantly alter underlying market dynamics.

Second, the absence of strong inductive biases in standard Transformers may limit their adaptability to noisy financial data. Several researchers within the machine learning community have argued that simpler forecasting methods sometimes outperform Transformers in highly stochastic environments.

Third, computational complexity remains a practical challenge. Transformer architectures require significant computational resources and large datasets for effective training. This may limit their accessibility in smaller financial institutions or low-latency trading systems.

The findings suggest that future research should focus on:

- Regime-aware attention mechanisms
- Adaptive volatility gating
- Hybrid econometric-deep learning systems
- Robust temporal encoding methods
- Interpretable Transformer architectures

Integrating financial domain knowledge into Transformer design may also improve model stability and interpretability under extreme market conditions.

6. Conclusion

This paper evaluated the robustness of Transformer-based time-series models for short-term financial volatility prediction across varying market regimes. Experimental findings demonstrated that Transformer architectures outperform several classical forecasting approaches in stable market environments by effectively modeling long-range temporal dependencies and nonlinear interactions.

However, model robustness decreases during abrupt market transitions and crisis-driven volatility spikes. Among the evaluated architectures, PatchTST and ETSformer demonstrated stronger adaptability and cross-regime stability compared with standard Transformer implementations.

The results indicate that while Transformer-based forecasting systems hold substantial promise for quantitative finance, additional research is necessary to improve robustness under non-stationary and adversarial market conditions. Future volatility forecasting systems should incorporate regime-awareness, adaptive learning mechanisms, and financial inductive biases to achieve reliable real-world deployment.

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